

BASIC LOGIC BUILDING-1

1. Electricity Bill Calculator

Difficulty: Easy
Time Limit: 15 minutes

Problem Statement
A local electricity board calculates the monthly electricity bill based on the number of units consumed. The billing system follows these rules:

- First 100 units → ₹5 per unit
- Next 100 units (101–200) → ₹7 per unit
- Next 200 units (201–400) → ₹10 per unit
- Above 400 units → ₹15 per unit

For example, if a customer consumes 250 units, the first 100 units are charged at ₹5, the next 100 at ₹7, and the remaining 50 at ₹10.
Write a program to calculate the total electricity bill.

Input Format
A single integer N, representing the number of units consumed.

Output Format
Print the total electricity bill.

Constraints

- $0 \le N \le 10,000$

Example

Input	**Output**
250	1700

Explanation

- First 100 units = 100 × 5 = ₹500
- Next 100 units = 100 × 7 = ₹700
- Remaining 50 units = 50 × 10 = ₹500

Total = ₹1700

Test Cases

Input: 80	Output: 400
Input: 150	Output: 850
Input: 500	Output: 4700



2. Number Classification

Difficulty: Easy
Estimated Time Limit: 15 minutes

Problem Statement
A mathematics teacher wants to analyze a positive integer entered by a student.
For the given number, determine whether it is:

- Even or Odd
- Positive or Zero
- Divisible by 5 or not

The program must display all applicable classifications.

Input Format
A single integer N.

Output Format
Print three lines:
Even/Odd
Positive/Zero
Divisible by 5/Not Divisible by 5

Constraints

- 0 ≤ N ≤ 10^9

Example

Input	**Output**
120	Even Positive Divisible by 5

Test Cases

Input: 17	Output: Odd Positive Not Divisible by 5
Input: 0	Output: Even Zero Divisible by 5



3. Digital Sum and Product

Difficulty: Easy
Estimated Time Limit: 15 minutes

Problem Statement
A number-analysis application needs to process the digits of a given positive integer.
Your task is to calculate:

1. The sum of all digits.
2. The product of all digits.
3. Whether the number is divisible by 3 based on its digit sum.

The program must process the number digit by digit using a loop.

Input Format
A single positive integer N.

Output Format
Print:
Sum: <sum>
Product: <product>
Divisible by 3: Yes/No

Constraints

- 1 ≤ N ≤ 10^9

Example

Input	**Output**
234	Sum: 9 Product: 24 Divisible by 3: Yes

Explanation
Digits are 2, 3, 4.
Sum:
2 + 3 + 4 = 9
Product:
2 × 3 × 4 = 24
Since 9 is divisible by 3, the number is divisible by 3.

Test Cases

Input: 12345	Output: Sum: 15

	Product: 120 Divisible by 3: Yes
Input: 124	Output: Sum: 7 Product: 8 Divisible by 3: No

4. Student Result Analyzer

Difficulty: Easy
Estimated Time Limit: 20 minutes

Problem Statement
A college wants to automatically generate results for students.
Each student has marks in N subjects. A student passes a subject only if the marks are at least 40.
The student passes overall only when they pass **every subject** and their average marks are at least 50.
Write a program to calculate the total marks, average marks, number of failed subjects, and final result.

Input Format
First line contains integer N, the number of subjects.
Second line contains N integers representing marks.

Output Format
Print:
Total: <total>
Average: <average>
Failed Subjects: <count>
Result: PASS/FAIL
Print the average up to **2 decimal places**.

Constraints
- $1 \le N \le 20$
- $0 \le$ marks ≤ 100

Example

Input	**Output**
5 65 72 55 80 68	Total: 340 Average: 68.00 Failed Subjects: 0 Result: PASS



Explanation

Total:

65 + 72 + 55 + 80 + 68 = 340

Average:

340 / 5 = 68

No subject has marks below 40, and the average is above 50.

Test Cases

Input: 4 80 75 90 85	Output: Total: 330 Average: 82.50 Failed Subjects: 0 Result: PASS
Input: 5 80 35 70 65 75	Output: Total: 325 Average: 65.00 Failed Subjects: 1 Result: FAIL

5. ATM Transaction Simulator

Difficulty: Medium

Estimated Time Limit: 25 minutes

Scenario

A bank wants to create a simple ATM simulation. The ATM starts with a particular account balance.

The customer can perform multiple transactions. For every transaction, the customer enters an operation:

- D → Deposit
- W → Withdraw
- B → Check Balance
- E → Exit

For withdrawal, the amount must not exceed the current balance. Invalid withdrawal requests must not change the balance.

The program should continue accepting transactions until the customer selects E.

Input Format

First line: initial balance.

Then multiple lines containing:

Operation Amount

For B and E, no amount is provided.



Output Format

For every transaction, print the corresponding result.
At the end print:
Final Balance: <balance>

Constraints

- 0 ≤ initial balance ≤ 10^6
- 0 < transaction amount ≤ 10^6
- Maximum 100 transactions

Example

Input	**Output**
5000 D 2000 W 1500 W 7000 B E	Deposit Successful Withdrawal Successful Insufficient Balance Balance: 5500 Final Balance: 5500

Explanation

Initial balance = 5000
After deposit:
5000 + 2000 = 7000
After withdrawal:
7000 - 1500 = 5500
Withdrawal of 7000 is rejected because the balance is only 5500.

6. Parking Fee Calculator

Difficulty: Medium
Estimated Time Limit: 25 minutes

Problem Statement

A shopping mall wants to automate its parking system.
The parking fee depends on the vehicle type:

- C — Car
- B — Bike
- T — Truck

The charges are:

Car

- First 2 hours → ₹30/hour
- Additional hours → ₹20/hour

Bike

- First 2 hours → ₹15/hour
- Additional hours → ₹10/hour

Truck

- First 2 hours → ₹50/hour
- Additional hours → ₹40/hour

Given multiple vehicles, calculate the total parking collection.

Input Format

First line contains integer N, the number of vehicles.
Next N lines contain:
VehicleType Hours

Output Format

Print:
Total Collection: <amount>

Constraints

- 1 ≤ N ≤ 1000
- 1 ≤ Hours ≤ 100

Example

Input	**Output**
3 C 3 B 2 T 4	Total Collection: 290

Explanation

Car:
2 × 30 + 1 × 20 = 80
Bike:
2 × 15 = 30
Truck:
2 × 50 + 2 × 40 = 180
Total:
80 + 30 + 180 = 290

Test Cases

Input:	Output:
2 C 2 B 5	Total Collection: 90

Input: 4 T 1 C 4 B 3 C 2	Output: Total Collection: 210

7. Smart Grocery Checkout

Difficulty: Medium
Estimated Time Limit: 30 minutes

Problem Statement

A supermarket wants to implement a smart billing system.
A customer purchases N products. For each product, the program receives its price and quantity. The system calculates the subtotal.
The supermarket provides discounts based on the subtotal:

- Below ₹1,000 → No discount
- ₹1,000–₹4,999 → 5% discount
- ₹5,000–₹9,999 → 10% discount
- ₹10,000 or more → 15% discount

After applying the discount, an additional 5% tax is charged on the discounted amount.

Input Format

First line contains N.
Next N lines contain:
Price Quantity

Output Format

Print:
Subtotal: <amount>
Discount: <amount>
Tax: <amount>
Final Amount: <amount>
Print monetary values up to 2 decimal places.

Constraints

- 1 ≤ N ≤ 100
- 1 ≤ Price ≤ 100000
- 1 ≤ Quantity ≤ 100

Example



Input	**Output**
3 500 2 1000 1 200 5	Subtotal: 3000.00 Discount: 150.00 Tax: 142.50 Final Amount: 2992.50

Explanation
Subtotal:
500×2 + 1000×1 + 200×5 = 3000
₹3000 falls into the 5% discount category.
Discount:
3000 × 5% = 150
Discounted amount:
3000 - 150 = 2850
Tax:
2850 × 5% = 142.50
Final amount:
2992.50

Test Cases

Input: 2 500 1 400 1	Output: Subtotal: 900.00 Discount: 0.00 Tax: 45.00 Final Amount: 945.00
Input: 2 5000 1 6000 1	Output: Subtotal: 11000.00 Discount: 1650.00 Tax: 467.50 Final Amount: 9817.50

8. Bank Loan Eligibility & EMI Schedule

Difficulty: Hard
Estimated Time Limit: 35 minutes

Problem Statement
A bank wants to automate its preliminary loan eligibility system.
A customer provides their monthly salary, existing monthly EMI, credit score, and requested loan amount.
The customer is eligible only if:

- Salary is at least ₹25,000.
- Credit score is at least 700.
- Existing EMI is not more than 40% of monthly salary.

If eligible, the bank determines the maximum permissible EMI as 50% of the salary minus the existing EMI. The program must then calculate a simple monthly repayment schedule using the given annual interest rate and number of months.

Input Format
Input contains:
Salary ExistingEMI CreditScore LoanAmount AnnualInterest Months

Output Format
If not eligible:
Loan Status: NOT ELIGIBLE
Otherwise print:
Loan Status: ELIGIBLE
Maximum EMI: <amount>
Estimated Monthly EMI: <amount>

Constraints

- 25000 ≤ Salary ≤ 1000000
- 0 ≤ ExistingEMI ≤ Salary
- 300 ≤ CreditScore ≤ 900
- 10000 ≤ LoanAmount ≤ 10000000
- 1 ≤ AnnualInterest ≤ 30
- 1 ≤ Months ≤ 120

Example

Input	**Output**
60000 10000 750 500000 12 24	Loan Status: ELIGIBLE Maximum EMI: 20000.00 Estimated Monthly EMI: 23536.74

Explanation
Salary = ₹60,000
Existing EMI ratio:
10000 / 60000 × 100 = 16.67%
Credit score = 750, which satisfies the requirement.
Maximum permissible EMI:
50% of 60000 - 10000 = 20000
The program then calculates the EMI for ₹5,00,000 over 24 months at 12% annual interest.
If the estimated EMI exceeds the maximum permissible EMI, the program should report that the requested loan amount cannot be approved under the EMI condition.



9. Traffic Signal Simulation

Difficulty: Hard
Estimated Time Limit: 40 minutes

Scenario

A city traffic department wants to simulate vehicles passing through a traffic signal.
The signal operates in three states:

- R → Red
- Y → Yellow
- G → Green

During a red signal, vehicles must stop. During yellow, vehicles slow down, and during green, vehicles can pass.
For each signal cycle, the system receives the signal and number of vehicles waiting. The number of vehicles that can pass depends on the signal:

- Red → 0 vehicles
- Yellow → 2 vehicles
- Green → 10 vehicles

If more vehicles are waiting than the allowed number, the remaining vehicles stay in the queue for the next cycle.

Input Format

First line contains N, number of signal cycles.
Next N lines contain:
Signal VehiclesWaiting

Output Format

For every cycle print:
Cycle i: Passed = X, Remaining = Y
At the end print:
Total Passed: X
Final Queue: Y

Constraints

- $1 \le N \le 1000$
- $0 \le$ VehiclesWaiting ≤ 100000
- Signal is one of R, Y, G

Example

Input	Output
4 G 8 R 5	Cycle 1: Passed = 8, Remaining = 0 Cycle 2: Passed = 0, Remaining = 5 Cycle 3: Passed = 2, Remaining = 5

Y 7 G 12	Cycle 4: Passed = 10, Remaining = 7 Total Passed: 20 Final Queue: 7

Explanation
Cycle 1:
Green allows 10 vehicles, but only 8 are waiting.
Passed = 8, Remaining = 0.
Cycle 2:
Red allows no vehicles.
Passed = 0, Remaining = 5.
Cycle 3:
7 vehicles are waiting. Yellow allows 2.
Passed = 2, Remaining = 5.
Cycle 4:
5 previous vehicles + 12 new vehicles = 17.
Green allows 10.
Passed = 10, Remaining = 7.

10. Cricket Tournament Score Analyzer

Difficulty: Hard
Estimated Time Limit: 45 minutes

Scenario
A college is conducting a cricket tournament. Each team plays multiple matches, and the organizers want to generate a basic performance report.
For every match, the program receives the runs scored by the team and the runs scored by the opponent.
The result is determined as follows:

- Team score > opponent score → Win
- Team score < opponent score → Loss
- Equal scores → Tie

The tournament awards:

- Win → 2 points
- Tie → 1 point
- Loss → 0 points

The program must process all matches and calculate total wins, losses, ties, points, total runs scored, and highest score.

Input Format
First line contains integer N, the number of matches.
Next N lines contain:

TeamRuns OpponentRuns

Output Format

Print:
Wins: <wins>p
Losses: <losses>
Ties: <ties>
Points: <points>
Total Runs: <runs>
Highest Score: <score>

Constraints

- 1 ≤ N ≤ 1000
- 0 ≤ TeamRuns, OpponentRuns ≤ 1000

Example

Input	**Output**
5 180 150 120 120 95 110 210 200 160 170	Wins: 2 Losses: 2 Ties: 1 Points: 5 Total Runs: 765 Highest Score: 210

Explanation

Match 1:
180 > 150 → Win → 2 points
Match 2:
120 = 120 → Tie → 1 point
Match 3:
95 < 110 → Loss → 0 points
Match 4:
210 > 200 → Win → 2 points
Match 5:
160 < 170 → Loss → 0 points
Therefore:

- Wins = 2
- Losses = 2
- Ties = 1
- Points = 5
- Total Runs = 765
- Highest Score = 210



Test Cases

Input: 3 100 90 120 130 150 150	Output: Wins: 1 Losses: 1 Ties: 1 Points: 3 Total Runs: 370 Highest Score: 150
Input: 4 200 100 250 180 175 120 300 299	Output: Wins: 4 Losses: 0 Ties: 0 Points: 8 Total Runs: 925 Highest Score: 300